

## CHAPTER 1: LARGE NUMBERS

## Comprehensive Exercise Answer Key

This document contains verified step-by-step solutions for all exercises, workouts, and mathematical lab activities in Chapter 1: Large Numbers.

## Exercise 1A — Number Names (Indian Place Value System)

## Q1. Match the following numbers with their number names:

| Number      | Matched Name                                              | Explanation / Indian System Place Value                                                                                |
|-------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| a) 73,604   | iv) Seventy-three thousand six hundred four               | 73,604 is written as Seventy-three thousand six hundred four. Commas separate the Thousands and Ones periods (73,604). |
| b) 9,96,400 | v) Nine lakh ninety-six thousand four hundred             | 9,96,400 has Lakhs (9), Thousands (96), and Ones (400). Read as Nine lakh ninety-six thousand four hundred.            |
| c) 40,063   | i) Forty thousand sixty-three                             | 40,063 has Thousands (40) and Ones (063 -> sixty-three). Read as Forty thousand sixty-three.                           |
| d) 2,57,189 | ii) Two lakh fifty-seven thousand one hundred eighty-nine | 2,57,189 has Lakhs (2), Thousands (57), and Ones (189). Read as Two lakh fifty-seven thousand one hundred eighty-nine. |
| e) 8,05,701 | iii) Eight lakh five thousand seven hundred one           | 8,05,701 has Lakhs (8), Thousands (05), and Ones (701). Read as Eight lakh five thousand seven hundred one.            |

## Exercise 1B — Successor, Predecessor, and Comparison

## Q1 &amp; Q2. Successors and Predecessors:

Rules: Successor of a number is  $\text{Number} + 1$ . Predecessor of a number is  $\text{Number} - 1$ .

| Original Number | Q1. Successor (+1) | Original Number | Q2. Predecessor (-1) |
|-----------------|--------------------|-----------------|----------------------|
| a) 72,086       | 72,087             | a) 82,761       | 82,760               |
| b) 80,209       | 80,210             | b) 84,510       | 84,509               |
| c) 36,099       | 36,100             | c) 11,000       | 10,999               |

| Original Number | Q1. Successor (+1) | Original Number | Q2. Predecessor (-1) |
|-----------------|--------------------|-----------------|----------------------|
| d) 5,57,299     | <b>5,57,300</b>    | d) 5,83,497     | <b>5,83,496</b>      |
| e) 4,99,360     | <b>4,99,361</b>    | e) 7,95,299     | <b>7,95,298</b>      |
| f) 8,14,999     | <b>8,15,000</b>    | f) 2,19,300     | <b>2,19,299</b>      |

### Q3. Compare and fill in the blanks with > or <:

Rule: Compare digit by digit starting from the highest place value.

| Expression                | Comparison Symbol          | Mathematical Reason                                   |
|---------------------------|----------------------------|-------------------------------------------------------|
| a) 7,183 ____ 7,456       | <b>&lt; (Less than)</b>    | At Hundreds place, 1 < 4. Thus, 7,183 < 7,456.        |
| b) 65,284 ____ 69,983     | <b>&lt; (Less than)</b>    | At Thousands place, 5 < 9. Thus, 65,284 < 69,983.     |
| c) 31,054 ____ 27,453     | <b>&gt; (Greater than)</b> | At Ten thousands place, 3 > 2. Thus, 31,054 > 27,453. |
| d) 7,52,572 ____ 7,52,496 | <b>&gt; (Greater than)</b> | At Hundreds place, 5 > 4. Thus, 7,52,572 > 7,52,496.  |
| e) 3,71,415 ____ 3,70,657 | <b>&gt; (Greater than)</b> | At Thousands place, 1 > 0. Thus, 3,71,415 > 3,70,657. |
| f) 8,23,414 ____ 8,23,417 | <b>&lt; (Less than)</b>    | At Ones place, 4 < 7. Thus, 8,23,414 < 8,23,417.      |

**Q4. Write the following numbers in ascending order (smallest to greatest):**

a) 45,382; 6,29,483; 5,73,132; 7,68,542

**Answer: 45,382 < 5,73,132 < 7,68,542 < 6,29,483**

b) 24,791; 37,832; 13,475; 8,71,043

**Answer: 13,475 < 24,791 < 37,832 < 8,71,043**

c) 12,461; 26,731; 26,715; 2,11,359

**Answer: 12,461 < 26,715 < 26,731 < 2,11,359**

d) 6,71,045; 5,83,714; 8,79,163; 4,72,819

**Answer: 4,72,819 < 5,83,714 < 6,71,045 < 8,79,163****Q5. Write the following numbers in descending order (greatest to smallest):**

a) 5,473; 6,74,831; 47,389; 68,432

**Answer: 6,74,831 > 68,432 > 47,389 > 5,473**

b) 99,879; 5,01,023; 43,765; 7,83,000

**Answer: 7,83,000 > 5,01,023 > 99,879 > 43,765**

c) 2,73,846; 9,87,876; 26,734; 3,46,712

**Answer: 9,87,876 > 3,46,712 > 2,73,846 > 26,734**

d) 9,01,126; 1,02,341; 9,45,720; 9,97,134

**Answer: 9,97,134 > 9,45,720 > 9,01,126 > 1,02,341****Exercise 1C — Number Formation****Q1. Form the greatest and smallest numbers with given digits (use each digit once):***Note: A number cannot begin with 0. If 0 is present, the second-smallest digit is placed first, followed by 0.*

| Given Digits     | Smallest Number | Greatest Number | Explanation                                             |
|------------------|-----------------|-----------------|---------------------------------------------------------|
| a) 6, 2, 8, 1, 5 | <b>12,568</b>   | <b>86,521</b>   | Digits in ascending/descending order.                   |
| b) 3, 0, 5, 4, 1 | <b>10,345</b>   | <b>54,310</b>   | Smallest cannot start with 0, so 1 comes first.         |
| c) 4, 9, 7, 6, 2 | <b>24,679</b>   | <b>97,642</b>   | Digits in ascending/descending order.                   |
| d) 2, 6, 4, 0, 9 | <b>20,469</b>   | <b>96,420</b>   | Smallest starts with 2, then 0. Greatest in descending. |
| e) 1, 0, 3, 7, 5 | <b>10,357</b>   | <b>75,310</b>   | Smallest starts with 1, then 0. Greatest in descending. |

## Q2. Write greatest and smallest 6-digit numbers using given digits (repeat digits, use each at least once):

*Strategy: To form the smallest number, repeat the smallest digit at the highest place values. To form the greatest number, repeat the greatest digit at the highest place values.*

| Given Digits | Smallest 6-digit Number | Greatest 6-digit Number | Mathematical Logic                                                          |
|--------------|-------------------------|-------------------------|-----------------------------------------------------------------------------|
| a) 3, 7, 1   | <b>1,11,137</b>         | <b>7,77,731</b>         | Smallest: Repeat '1' 4 times. Greatest: Repeat '7' 4 times.                 |
| b) 5, 0, 6   | <b>5,00,006</b>         | <b>6,66,650</b>         | Smallest starts with '5', followed by repeated '0's. Greatest repeats '6's. |
| c) 0, 8, 4   | <b>4,00,008</b>         | <b>8,88,840</b>         | Smallest starts with '4', followed by repeated '0's. Greatest repeats '8's. |

## Exercise 1D — Rounding Off Numbers

*Rules: To round off to nearest Ten, check Ones digit ( $\geq 5$  rounds UP, else DOWN). For Hundred, check Tens digit ( $\geq 50$  rounds UP). For Thousand, check Hundreds digit ( $\geq 500$  rounds UP).*

| No. | Number                                 | Q1. Nearest Ten                       | Q2. Nearest Hundred                   | Q3. Nearest Thousand                  |
|-----|----------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| a   | 54 (a) / 673 (b) / 3,923 (c)           | <b>50</b> (Ones digit 4 < 5)          | <b>700</b> (Tens digit 7 $\geq$ 5)    | <b>4,000</b> (Hundreds 9 $\geq$ 5)    |
| b   | 387 (a) / 5,173 (b) / 6,456 (c)        | <b>390</b> (Ones digit 7 $\geq$ 5)    | <b>5,200</b> (Tens digit 7 $\geq$ 5)  | <b>6,000</b> (Hundreds 4 < 5)         |
| c   | 12,961 (a) / 4,721 (b) / 52,198 (c)    | <b>12,960</b> (Ones digit 1 < 5)      | <b>4,700</b> (Tens digit 2 < 5)       | <b>52,000</b> (Hundreds 1 < 5)        |
| d   | 67,115 (a) / 32,805 (b) / 1,37,805 (c) | <b>67,120</b> (Ones digit 5 $\geq$ 5) | <b>32,800</b> (Tens digit 0 < 5)      | <b>1,38,000</b> (Hundreds 8 $\geq$ 5) |
| e   | 7,407 (a) / 39,166 (b) / 59,566 (c)    | <b>7,410</b> (Ones digit 7 $\geq$ 5)  | <b>39,200</b> (Tens digit 6 $\geq$ 5) | <b>60,000</b> (Hundreds 5 $\geq$ 5)   |

## Exercise 1E — International Place Value System

### Q1. Place commas according to the International system and write number names:

*Note: Periods are divided into groups of 3 (Ones, Thousands, Millions). Commas are placed after every 3 digits from the right.*

| No. | Raw   | With Commas | International Number Name                     |
|-----|-------|-------------|-----------------------------------------------|
| a   | 29175 | 29,175      | Twenty-nine thousand one hundred seventy-five |

| No. | Raw    | With Commas | International Number Name                                  |
|-----|--------|-------------|------------------------------------------------------------|
| b   | 72083  | 72,083      | Seventy-two thousand eighty-three                          |
| c   | 134867 | 134,867     | One hundred thirty-four thousand eight hundred sixty-seven |
| d   | 370617 | 370,617     | Three hundred seventy thousand six hundred seventeen       |
| e   | 717305 | 717,305     | Seven hundred seventeen thousand three hundred five        |
| f   | 705260 | 705,260     | Seven hundred five thousand two hundred sixty              |
| g   | 401729 | 401,729     | Four hundred one thousand seven hundred twenty-nine        |
| h   | 742003 | 742,003     | Seven hundred forty-two thousand three                     |
| i   | 471030 | 471,030     | Four hundred seventy-one thousand thirty                   |
| j   | 202889 | 202,889     | Two hundred two thousand eight hundred eighty-nine         |

### Q2. Write numbers for the given names using International system:

- a) Five hundred eighty-two thousand two hundred → **582,200**  
 b) Four hundred forty-eight thousand two hundred twelve → **448,212**  
 c) Six hundred forty-five thousand four hundred five → **645,405**  
 d) Eight hundred fifty-seven thousand six hundred three → **857,603**  
 e) Two hundred thousand seven hundred fifty → **200,750**

## Exercise 1F — Roman Numerals

### Q1 & Q2. Roman Numerals Writing and Conversion:

Rules: I=1, V=5, X=10, L=50, C=100, D=500, M=1,000. Repeat up to 3 times. Small left of large means subtraction; right means addition.

| Hindu-Arabic | Q1. Roman Numeral              | Roman Numeral | Q2. Hindu-Arabic   |
|--------------|--------------------------------|---------------|--------------------|
| a) 14        | XIV (10 + 5 - 1)               | a) XXVIII     | 28 (10+10+5+1+1+1) |
| b) 78        | LXXVIII (50 + 10 + 10 + 5 + 3) | b) VII        | 7 (5+1+1)          |
| c) 54        | LIV (50 + 5 - 1)               | c) LXXV       | 75 (50+10+10+5)    |
| d) 129       | CXXIX (100 + 20 + 9)           | d) XXII       | 22 (10+10+1+1)     |
| e) 45        | XLV (50 - 10 + 5)              | e) XCII       | 92 (100 - 10 + 2)  |
| f) 282       | CCLXXXII (200 + 80 + 2)        | f) XCIX       | 99 (100-10 + 10-1) |
| g) 313       | CCCXIII (300 + 10 + 3)         | g) IX         | 9 (10 - 1)         |

| Hindu-Arabic | Q1. Roman Numeral | Roman Numeral | Q2. Hindu-Arabic |
|--------------|-------------------|---------------|------------------|
| h) 28        | XXVIII (20 + 8)   | h) LVI        | 56 (50 + 5 + 1)  |

### Q3. Multiplication tables in Roman numerals:

- a)  $2 \times 5 = 10 \rightarrow \text{X}$   
 b)  $6 \times 8 = 48 \rightarrow \text{XLVIII}$   
 c)  $3 \times 9 = 27 \rightarrow \text{XXVII}$   
 d)  $7 \times 7 = 49 \rightarrow \text{XLIX}$   
 e)  $4 \times 7 = 28 \rightarrow \text{XXVIII}$   
 f)  $5 \times 6 = 30 \rightarrow \text{XXX}$

## Chapter Review: Workout (Pages 16-17)

### Q1. Write the face value and place value of the colored digits:

*Note: Face value of a digit is the digit itself. Place value depends on its position.*

| Number      | Colored Digit (Red) | Face Value | Place Value | Indian Position |
|-------------|---------------------|------------|-------------|-----------------|
| a) 27,854   | 7                   | 7          | 7,000       | Thousands (Th)  |
| b) 53,062   | 3                   | 3          | 3,000       | Thousands (Th)  |
| c) 4,90,820 | 0                   | 0          | 0           | Thousands (Th)  |
| d) 6,30,575 | 5                   | 5          | 500         | Hundreds (H)    |
| e) 2,05,127 | 5                   | 5          | 5,000       | Thousands (Th)  |

**Q2. Write the following numbers in expanded form:**

- a) 120458 → **1,00,000 + 20,000 + 400 + 50 + 8** (1 lakh + 2 ten thousands + 4 hundreds + 5 tens + 8 ones)  
b) 368405 → **3,00,000 + 60,000 + 8,000 + 400 + 5**  
c) 705230 → **7,00,000 + 5,000 + 200 + 30**  
d) 848007 → **8,00,000 + 40,000 + 8,000 + 7**  
e) 241625 → **2,00,000 + 40,000 + 1,000 + 600 + 20 + 5**

**Q3. Write the standard form of the following numbers:**

- a) 20000 + 5000 + 400 + 8 → **25,408** (no tens digit, so '0' is placed in Tens place)  
b) 30000 + 700 + 60 + 5 → **30,765** (no thousands digit)  
c) 400000 + 6000 + 90 + 9 → **4,06,099** (no ten thousands or hundreds digits)  
d) 900000 + 20000 + 3000 + 70 → **9,23,070** (no hundreds or ones digits)

**Q4. Circle the smallest number and underline the greatest number:**

- a) 39,872; 54,781; 22,780; 60,708 → Smallest: **22,780** (Circled), Greatest: **60,708** (Underlined)  
b) 7,53,281; 7,53,218; 7,53,382; 7,53,177 → Smallest: **7,53,177** (Circled), Greatest: **7,53,382** (Underlined)

**Q5. Write the numbers in descending order and write the name of the greatest in the Indian system:**

- a) 23,781; 5,70,966; 7,27,820; 5,78,325  
Descending Order: **7,27,820 > 5,78,325 > 5,70,966 > 23,781**  
Greatest Number: **7,27,820**  
Indian Number Name: **Seven lakh twenty-seven thousand eight hundred twenty**
- b) 9,21,345; 9,21,445; 9,21,159; 9,21,560  
Descending Order: **9,21,560 > 9,21,445 > 9,21,345 > 9,21,159**  
Greatest Number: **9,21,560**  
Indian Number Name: **Nine lakh twenty-one thousand five hundred sixty**

**Q6. Write the place value of digit 4 in the following:**

- a) 4,38,715 → **4,00,000** (Lakhs place)  
 b) 7,58,426 → **400** (Hundreds place)  
 c) 6,42,173 → **40,000** (Ten thousands place)  
 d) 8,94,157 → **4,000** (Thousands place)  
 e) 7,61,045 → **40** (Tens place)

**Q7. Add and express the sum in Roman numerals:**

- a)  $25 + 32 = 57$  → **LVII** ( $50 + 5 + 2$ )  
 b)  $102 + 12 = 114$  → **CXIV** ( $100 + 10 + 4$ )  
 c)  $77 + 19 = 96$  → **XCVI** ( $100 - 10 + 6$ )  
 d)  $59 + 96 = 155$  → **CLV** ( $100 + 50 + 5$ )

**Q8. Match the following numbers with their names (Workout Page 17):**

- a) 6,74,829 → **iii) Six lakh seventy-four thousand eight hundred twenty-nine**  
 b) 97,156 → **iv) Ninety-seven thousand one hundred fifty-six**  
 c) 3,45,281 → **ii) Three hundred forty-five thousand two hundred eighty-one**  
 d) 6,18,279 → **i) Six hundred eighteen thousand two hundred seventy-nine**

**Q9. Round off expenditures to the nearest Hundred:**

- a) Books = ₹ 573 → ₹ **600** (Tens digit is 7  $\geq$  5)  
 b) Uniforms = ₹ 624 → ₹ **600** (Tens digit is 2  $<$  5)  
 c) Stationery = ₹ 212 → ₹ **200** (Tens digit is 1  $<$  5)  
 d) Sports Shoes = ₹ 785 → ₹ **800** (Tens digit is 8  $\geq$  5)

**Q10. In number 3,52,671, write the place value of:**

- a) Digit 3 → **3,00,000** (Lakhs place)  
 b) Digit 1 → **1** (Ones place)

**Exercise 1H — Perform & HOTS (Page 18)****Q1. Perform Rounding Off Table:**

| Number   | Nearest Ten  | Nearest Hundred | Nearest Thousand |
|----------|--------------|-----------------|------------------|
| a) 1,234 | <b>1,230</b> | <b>1,200</b>    | <b>1,000</b>     |



| Number    | Nearest Ten | Nearest Hundred | Nearest Thousand |
|-----------|-------------|-----------------|------------------|
| b) 75,392 | 75,390      | 75,400          | 75,000           |
| c) 66,039 | 66,040      | 66,000          | 66,000           |
| d) 89,989 | 89,990      | 90,000          | 90,000           |
| e) 10,005 | 10,010      | 10,000          | 10,000           |

### HOTS (Higher Order Thinking Skills):

**Question:** Which of the following data should never be considered for rounding off?

- a) Population of India
- b) Your age
- c) Your marks in Maths
- d) Flight departure time

**Answer:** d) Flight departure time

**Reasoning:** Rounding off population, age, or marks gives a close approximation that is useful and harmless in general conversation or statistics. However, *flight departure time* requires absolute precision. If a flight departs at 14:52 and we round it off to the nearest Ten (14:50) or Hour (15:00), it can cause travelers to miss their flight. Transport schedules, medicine dosages, and structural safety tolerances must never be rounded off in practice.

### Life Skills and Values conceptual question: Why are OTPs (One-Time Passwords) always 6-digit and not 7-digit numbers?

**Answer:** OTPs balance **security** and **usability** (human memory). A 6-digit number has 1,000,000 possible combinations (from 000000 to 999999). This makes it highly secure against automated hacking (brute-force) within its brief 30-to-120-second active window. Adding a 7th digit increases security only marginally, but significantly degrades user experience, making it harder for a user to quickly memorize, read off a phone screen, and type without errors.